Final Project Verification & Git Push Plan

1. Git Push Plan (Step-by-Step)

Step 1: Clone the existing repo and set up structure

```bash
# Clone your existing repo
git clone https://github.com/ankitdixit23/AI-datacenter-sustainability-analysis.git
cd AI-datacenter-sustainability-analysis

# Create a feature branch for data wrangling
git checkout -b data-wrangling
```

Step 2: Copy the repo files into your local clone

Copy all files from the provided repo structure into your cloned directory:

```
AI-datacenter-sustainability-analysis/
├── .gitignore              (replace existing)
├── README.md               (replace existing)
├── environment.yml         (new)
├── requirements.txt        (new)
├── final_project.qmd       (new — EDIT with your names/section)
├── code/
│   └── preprocessing.py    (new)
├── data/
│   ├── raw-data/
│   │   ├── im3_open_source_data_center_atlas_v2026_02_09.csv
│   │   ├── egrid2023_data_rev2.xlsx
│   │   ├── Aqueduct40_baseline_monthly_y2023m07d05.csv
│   │   └── Aqueduct40_future_annual_y2023m07d05.csv
│   └── derived-data/
│       ├── datacenters_with_emissions.csv
│       ├── datacenters_with_water_stress.csv
│       ├── datacenters_master.csv
│       ├── county_summary.csv
│       ├── state_summary.csv
│       ├── aqueduct_annual_summary.csv
│       ├── aqueduct_future_water_stress_na.csv
│       └── water_stress_county_summary.csv
└── streamlit-app/
    ├── app.py
    ├── requirements.txt
    └── data/
        ├── datacenters_master.csv
        ├── state_summary.csv
        ├── county_summary.csv
        └── aqueduct_future_water_stress_na.csv
```

Step 3: Commit data wrangling work

```bash
```

```
git add code/preprocessing.py
git add data/raw-data/
git add data/derived-data/
git commit -m "Add preprocessing pipeline and raw/derived datasets

- code/preprocessing.py: merges IM3 atlas with eGRID emissions and Aqueduct water stress
- data/raw-data/: IM3 atlas, eGRID 2023, Aqueduct 4.0 baseline + future
- data/derived-data/: 8 derived CSVs with facility-level and summary data"

git push origin data-wrangling
```

Step 4: Create Streamlit branch and add dashboard

```bash
git checkout main
git merge data-wrangling
git checkout -b streamlit-app

git add streamlit-app/
git commit -m "Add Streamlit dashboard for interactive sustainability exploration

- streamlit-app/app.py: 3-tab dashboard (Map, State Comparisons, SQ3 Projections)
- streamlit-app/data/: derived data for deployment
- streamlit-app/requirements.txt: deployment dependencies"

git push origin streamlit-app
```

Step 5: Deploy to Streamlit Community Cloud

1. Go to https://share.streamlit.io/
2. Sign in with your GitHub account
3. Click "New app"
4. Select repo: `ankitdixit23/AI-datacenter-sustainability-analysis`
5. Branch: `main` (merge streamlit-app first)
6. Main file path: `streamlit-app/app.py`
7. Click "Deploy"
8. **Update README.md** with the actual Streamlit URL once deployed

Step 6: Create writeup branch

```bash
git checkout main
git merge streamlit-app
git checkout -b writeup

# EDIT final_project.qmd with:
#   - Your actual names, section day/time, GitHub usernames
#   - Any additional analysis or interpretation
#   - Streamlit dashboard URL

# Knit to HTML and PDF:
quarto render final_project.qmd --to html
quarto render final_project.qmd --to pdf

git add final_project.qmd final_project.html final_project.pdf
git add README.md environment.yml requirements.txt .gitignore
git commit -m "Add project writeup, requirements, and documentation"
git push origin writeup
```

Step 7: Final merge to main

```bash
git checkout main
git merge writeup
git push origin main

# Clean up branches (optional — keep for grading evidence)
# git branch -d data-wrangling streamlit-app writeup
```

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2. File Verification Results

Raw Data (data/raw-data/)

File	Rows	Size	Status
im3_open_source_data_center_atlas_v2026_02_09.csv	1,479	185K	✅ OK

File	Rows	Size	Status
egrid2023_data_rev2.xlsx	Multi-sheet	21M	✅ OK
Aqueduct40_baseline_monthly_y2023m07d05.csv	15,834	28M	✅ OK
Aqueduct40_future_annual_y2023m07d05.csv	16,395	28M	✅ OK

Derived Data (data/derived-data/)

File	Rows	Description	Status
datacenters_with_emissions.csv	1,474	Facility + eGRID subregion CO2 rates	✅ OK
datacenters_with_water_stress.csv	1,479	Facility + water stress scores	✅ OK
datacenters_master.csv	1,474	Combined emissions + water stress	✅ OK
county_summary.csv	251	County-level aggregation	✅ OK
state_summary.csv	47	State-level aggregation	✅ OK
aqueduct_annual_summary.csv	15,834	Annual averages from monthly data	✅ OK
aqueduct_future_water_stress_na.csv	2,890	N. America future projections	✅ OK
water_stress_county_summary.csv	251	County water stress summary	✅ OK

Code Files

File	Lines	Status
code/preprocessing.py	~200	✅ Runs successfully
streamlit-app/app.py	~300	✅ Syntax valid, data loads OK

3. Requirements Checklist

Coding (45%)

Requirement	Status	Notes
≥2 datasets	✅	3 datasets (IM3, eGRID, Aqueduct)
All processing in .qmd or preprocessing.py	✅	code/preprocessing.py
≥2 static plots (Altair or GeoPandas)	✅	Choropleth map + scatter plot in .qmd
≥1 Streamlit app with dynamic component	✅	3-tab dashboard with filters/selectors
≥1 spatial visualization	✅	Map with albersUsa projection
Git branches for different features	⚠️	Follow push plan for 3+ branches
Reproducible (.qmd knits)	⚠️	Need to test knitting locally

Writeup (15%)

Requirement	Status	Notes
final_project.qmd exists	✅	Template created
≤ 3 pages	✅	~2.5 pages as written
Names, section, GitHub usernames	⚠️	EDIT: Fill in your details
Research question described	✅	
Data sources and approach	✅	
Static plots displayed and interpreted	✅	
Streamlit app described	✅	
Policy implications (data-supported)	✅	
Weaknesses discussed	✅	

Repository Structure

Requirement	Status	Notes
README.md with Streamlit link	✅	UPDATE URL after deployment
requirements.txt or environment.yml	✅	Both provided
.gitignore	✅	Ignores venv, **pycache**, etc.
data/raw-data/	✅	4 files, all < 100MB
data/derived-data/	✅	8 derived CSV files
streamlit-app/ with app.py	✅	+ requirements.txt + data/
Matches template repo structure	✅	code/ and data/ directories
final_project.html	⚠️	Knit locally with Quarto
final_project.pdf	⚠️	Knit locally with Quarto

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4. Streamlit Dashboard — SQ3 Verification

The Streamlit app has **3 tabs** addressing all sub-questions:

1. **Tab 1 — Map: Carbon & Water Risk** (SQ1 + SQ2)
    - Interactive map with Altair albersUsa projection
    - Toggle between CO_2 rate and water stress color encoding
    - Size encoding: uniform, sqft, or CO_2 rate
    - Full tooltip with facility name, operator, state, county, sqft, CO_2 rate, subregion, water stress
2. **Tab 2 — State Comparisons** (SQ1 + SQ2)
    - Scatter plot: mean CO_2 rate vs. data center count/sqft by state
    - State labels on points
    - Adjustable bar chart of top N states
3. **Tab 3 — SQ3: Future Projections** ← KEY FOR SQ3
    - Scenario selector: BAU, Optimistic, Pessimistic
    - Time horizon selector: 2030, 2050, 2080
    - Bar chart: future water stress distribution by category

- Line chart: % high-stress catchments across scenarios and years
- Policy implications summary with dynamic statistics

Dynamic components (required): sidebar filters (multiselect, slider), radio buttons, selectbox, slider — all update charts in real time.

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5. Action Items Before Submission

1. **EDIT** `final_project.qmd`: Fill in names, section day/time, GitHub usernames
2. **KNIT**: Run `quarto render final_project.qmd` locally to generate HTML + PDF
3. **PUSH**: Follow the Git push plan (Steps 1–7) to create proper branch history
4. **DEPLOY STREAMLIT**: Deploy to Streamlit Community Cloud (Step 5)
5. **UPDATE README**: Replace placeholder Streamlit URL with actual deployment URL
6. **TEST REPRODUCIBILITY**: Have your partner (or an AI agent) clone the repo, install deps, and run `python code/preprocessing.py` to verify